

DisasterClaw: A georeferenced simulation and agent framework for budgeted UAV disaster inspection

Anonymous authors

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Abstract

Aerial disaster inspection requires deciding whether another observation will help answer the current question, not merely whether it will sharpen a damage prediction. We present DisasterClaw, a georeferenced imagery-grounded simulation and agent framework that links movement, rendered observations, binary building-damage perception, and terminal question answering in traceable episodes. A development-set camera-ladder diagnostic shows that a narrower footprint can both correct and harm building predictions. We then evaluate seven frozen online policies on 160 questions independently reviewed by two annotators from 44 regions in three disaster events, with three generation repeats (3,360 episodes). The task-conditioned policy attains 75.00% accuracy, compared with 72.29% for no reobservation, while executing 161 reobservations compared with 923 for the always-reobserve policy. Its observed gain over no reobservation is 2.71 percentage points, but the prespecified ROI-cluster, multiplicity-adjusted interval spans -2.04 to $+7.50$ points; the experiment therefore does not establish a stable advantage over holding. The results support a platform and measurement contribution: the value of an additional observation depends on the task, and view changes can produce both beneficial and harmful answer transitions. Conclusions are limited to rendered orthographic imagery, simplified motion, a fixed ChangeOS backbone with prior xBD exposure, and three evaluation events.

Keywords: UAV disaster inspection, Embodied AI, Simulation platform, Active perception, Budgeted reobservation, Vision-language agents

1. Introduction

Disaster response creates a demanding setting for aerial embodied intelligence. An inspection agent must interpret a task, place its sensor over a relevant region, decide whether the current evidence is sufficient, and return a usable assessment under limited observation and motion budgets. Static

remote-sensing benchmarks measure recognition on fixed images, but they do not expose the decision of where and when to observe again. General aerial navigation environments provide motion and language tasks, yet they do not necessarily connect a changed view to a disaster-specific answer. A reproducible bridge between these settings is needed to study whether an agent’s sensing actions improve the decision that motivated them.

The bridge is technically nontrivial. Disaster imagery is often available as georeferenced, orthographic products rather than as a continuously rendered three-dimensional world. Apparent altitude changes must therefore preserve the source-resolution ceiling and maintain a common geographic reference. The agent also operates across several units of analysis: buildings are used by the damage model, regions define task scope, actions incur observation cost, and complete episodes are scored by final answers. Treating these units as interchangeable can make a classifier improvement appear to be an agent improvement. Generation variability adds another ambiguity when a vision-language model can change its answer without any environmental change.

We present DisasterClaw, a georeferenced simulation and agent framework for single-UAV disaster inspection. It transforms paired pre- and post-disaster imagery into an interactive observation environment with a map-anchored scene, a configurable camera footprint, a simplified vehicle state, finite movement and inspection actions, and episode logging. A disaster-response agent receives a natural-language task, a rendered view, and structured spatial evidence; it can search, center on evidence, descend to obtain a narrower footprint, answer, or abstain. The same environment supports an operator-facing console and a headless benchmark runner, so qualitative inspection and quantitative experiments use a shared world and action interface.

DisasterClaw is used here to examine budgeted reobservation. A closer view allocates more source pixels to the target region while discarding some surrounding context. Whether this helps depends on the task: a building label, a count, and a spatial relation can require different evidence. We first characterize the ChangeOS binary perception response across three footprints on a development population. The main experiment then compares frozen online policies on a separate, human-reviewed question set, scoring complete answers rather than classifier outputs alone.

Recent aerial embodied benchmarks emphasize broad skill coverage, realistic navigation, or standardized perception–decision–action evaluation, while recent remote-sensing agent benchmarks emphasize multimodal reasoning and tool use over large image collections. DisasterClaw targets a narrower complementary question: given a georeferenced disaster scene and a task-specific evidence requirement, can an executed observation action be traced

to a change in perception evidence and ultimately to a corrected or harmed terminal answer? The contribution is therefore not broader environment realism or a new foundation model, but a controlled measurement interface for task-dependent observation value in disaster inspection.

The contributions are as follows.

1. We develop a georeferenced aerial disaster-inspection environment and evaluation framework that converts paired disaster imagery into traceable closed-loop observation episodes. Unlike general aerial embodied benchmarks aimed at broad flight skills or simulator fidelity, the framework is designed to measure whether a controlled view change alters task-relevant evidence and the terminal disaster-inspection answer.
2. We implement a closed-loop inspection and measurement protocol that connects task language, rendered observations, a fixed binary ChangeOS detector, geographic evidence memory, movement tools, and Qwen2.5-VL-7B-Instruct answer generation. It records both beneficial and harmful answer changes and distinguishes executed observation cost from geometric motion estimates.
3. We compare seven frozen policies over 160 final questions and three generation repeats. The task-conditioned policy is numerically more accurate than holding, random reobservation, always reobserving, and raw entropy, while using fewer reobservations; the prespecified interval does not establish a stable advantage over holding or raw entropy. This mixed result is evidence about the task-dependent observation mechanism, not a general claim of policy superiority.

The present platform is an imagery-based research environment, not a six-degree-of-freedom flight simulator. It uses orthographic source imagery and simplified motion, and it does not model wind, aircraft dynamics, occlusion, collision avoidance, or battery discharge. These boundaries determine the scope of every experimental claim.

2. Related work

2.1. Aerial embodied agents and simulation platforms

Vision-and-language navigation connects natural-language instructions to sequential visual decisions [1]. AerialVLN extends this problem to UAV navigation, while CityNav introduces large-scale real-world aerial navigation data [2, 3]. OpenFly provides a broader aerial vision-language platform, toolchain, and trajectories across multiple heights and scenes [4]. These works establish language-conditioned aerial motion and height-varying observation as active research areas. DisasterClaw addresses a complementary requirement:

constructing task-scoped, paired disaster observations and measuring whether an inspection action changes the requested disaster assessment.

Large pretrained models can be composed with perception and navigation modules without treating the entire stack as a newly learned end-to-end policy [5]. CityNavAgent uses hierarchical semantic planning and global memory, spatial text representations support language-model reasoning in aerial navigation, and AerialClaw exposes modular aerial skills and feedback [6, 7, 8]. DisasterClaw adopts this modular view. Semantic maps, planning, memory, and tool execution support the implemented agent; the platform contribution lies in their integration with georeferenced disaster scenes, controllable inspection observations, and a common measurement interface.

CityEQA connects active urban exploration to embodied question answering and evaluates answers after view refinement [9]. Its setting is a direct precedent for active aerial question answering. DisasterClaw differs in task domain, source imagery, action abstraction, and evaluation target. Rather than benchmarking general-purpose flight skills or high-fidelity navigation, it couples a georeferenced disaster scene to a controlled observation transition and measures whether that transition changes task-relevant evidence and the terminal inspection answer. This narrower scope motivates reporting observation construction, action execution, and task-level correction or harm with equal precision.

Recent work has expanded aerial embodied intelligence beyond instruction-following navigation. AeroVerse organizes UAV-agent evaluation around scene awareness, spatial reasoning, navigational exploration, task planning, and motion decision, providing a broad benchmark suite for aerospace embodied world models [10]. BEDI further formalizes UAV embodied-agent evaluation through a perception–decision–action chain and evaluates semantic and spatial perception, motion control, tool use, planning, and action generation across real and virtual settings [11]. OpenUAV and the UAV-Need-Help benchmark emphasize realistic aerial trajectory execution and assistant-guided object search in three-dimensional environments [12]. These platforms address general aerial embodiment, realistic navigation, and standardized skill evaluation. DisasterClaw is complementary rather than competing in simulator fidelity: it uses georeferenced pre-/post-disaster imagery to make the task value of a controlled view change directly observable at the final disaster-inspection answer.

2.2. Disaster interpretation from remote-sensing imagery

xBD provides paired satellite images and building-damage annotations for disaster assessment [13]. RescueADI studies autonomous tool use for adaptive interpretation of remote-sensing disaster images [14]. Thus, paired

damage recognition and agentic disaster analysis both predate this work. Recent remote-sensing resources further broaden the evaluation of multimodal reasoning and agentic analysis. ThinkGeo evaluates tool-augmented language-model agents on multi-step remote-sensing tasks, including disaster assessment and change analysis, and scores both intermediate tool execution and final answers [15]. DisasterM3 provides a large-scale multi-hazard, multi-sensor vision-language benchmark for disaster damage assessment and response, covering perception and reasoning across diverse disaster events [16]. These resources strengthen static-image reasoning, multimodal generalization, and structured tool use. DisasterClaw addresses a different axis: the observation itself is an environment variable, so the agent can execute a view-changing action and the benchmark can measure whether the newly acquired evidence improves or harms the final task answer. DisasterClaw uses such static records as the substrate of an interactive inspection environment: a geographic window becomes an observation, vehicle state changes the window, and the resulting episode is scored at both building and task levels.

This transformation remains constrained by the data. Cropping and resampling an orthographic product can alter the number of pixels assigned to a target, but it cannot synthesize unrecorded viewpoints or detail below the native source scale. The platform is consequently suited to controlled studies of spatial coverage, observation allocation, and answer formation. Evidence from it cannot be presented as a measurement of real UAV image formation or field performance.

2.3. Active perception, uncertainty, and task value

Active perception treats sensing as an action selected in service of a task [17]. Planning under partial observability further connects information gathering with uncertain states and action outcomes [18]. This perspective motivates the central platform requirement: a new observation should be evaluated by the decision it enables and the cost it incurs.

Entropy reduction is a convenient trigger but is not equivalent to expected task utility. A distribution can become more concentrated around an incorrect class, while a view may help a count or spatial relation without changing the most uncertain building label. DisasterClaw therefore records classification, correction and harm, final question correctness, and executed observation count as separate endpoints.

Temperature scaling addresses probability calibration, while evaluation under dataset shift shows why in-distribution calibration should not be treated as a universal reliability property [19, 20]. Selective prediction measures the trade-off between coverage and error, and uncertainty-aware language-model planning has been studied through help-seeking behavior [21, 22]. We use

these distinctions to separate raw and calibrated binary-damage triggers, prediction-set sensing rules, and the VLM’s generated answer. None alone establishes reliability for the composed agent.

3. The DisasterClaw simulation platform

3.1. Scope and system organization

DisasterClaw is an imagery-based simulation and evaluation framework for single-UAV disaster inspection. Its purpose is to turn georeferenced disaster records into closed-loop episodes in which an agent changes its map position or observation footprint, receives a new image, and is scored on a disaster task. The implementation has four connected layers (Fig. 1): a task interface, a disaster-response agent, a simulated geospatial world, and an instrumentation layer. The operator-facing path uses a React/Vite console connected to a Flask/Socket.IO service. Quantitative experiments use a headless runner that calls the same environment and controller modules without rendering the user interface.

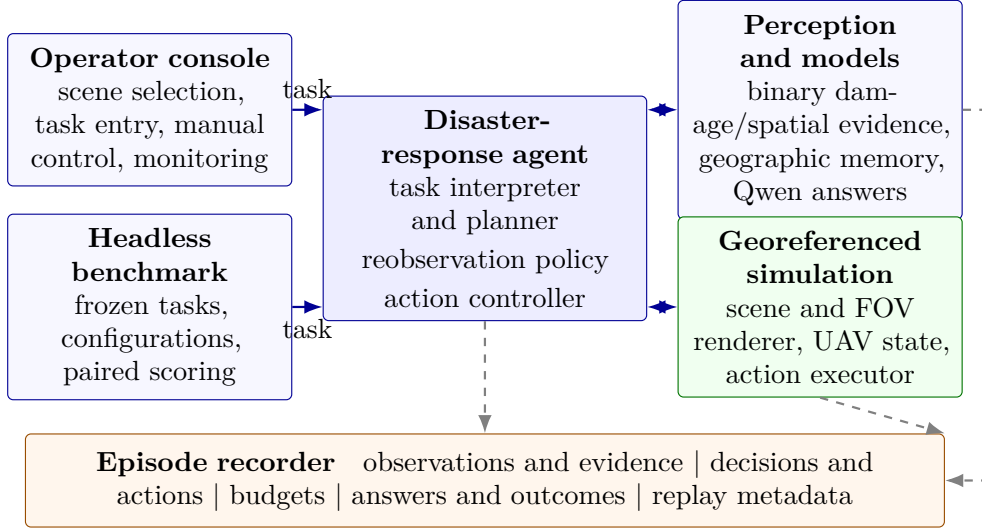


Figure 1: DisasterClaw system organization. The operator console and headless benchmark connect the disaster-response agent to the same perception/model services and georeferenced environment. The recorder retains intermediate model traces, environment transitions, budgets, and task outcomes.

Table 1 states what each layer implements and the boundary attached to it. This distinction is central to the intended use of the platform. Its geospatial state and image renderer support controlled observation studies; they do not reproduce the complete physics of an aircraft or camera.

Table 1: Implemented DisasterClaw capabilities and their evaluation boundaries.

Capability	Implemented function	Boundary in this study
Scene	Georeferenced pre-/post-disaster tiles, building annotations, map anchors, target ROIs, and contextual basemap	Three evaluated events and 44 ROIs; incomplete source archive and no global coverage claim
Observation	Nadir fixed-FOV crop and resampling at three footprint widths; paired archival and current-image channels	Orthographic imagery; no new viewpoint, acquisition time, or detail below the assumed 0.5 m/pixel source scale
Vehicle and actions	Geographic and local position, altitude, heading, speed, task state, straight-line motion, centering, descent, hover, marking, and reporting	One UAV with simplified kinematics; no six-degree-of-freedom dynamics, wind, collision, communication, or battery model
Perception and models	Building localization and binary no-damage/damaged evidence, geographic cross-view evidence memory, task-specific spatial summaries, and Qwen answer generation	Fixed ChangeOS backend with prior xBD exposure; fixed Qwen2.5-VL-7B-Instruct answer model; no claim of event-disjoint perception generalization
Tasks	Operator instructions, damage/presence/count/spatial questions, and archived language-guided navigation instructions	Task-scoped simulation; no claim of unrestricted natural-language or open-world competence
Logging and evaluation	Interactive monitoring, headless episodes, paired task IDs, intermediate evidence, actions, budgets, answers, and terminal scores	All 3,360 final online episodes are covered by local shard-level consistency audits; complete public release packaging remains pending
Physical boundary	Map-anchored scene transitions and a controlled observation ladder	No photorealistic sensor synthesis, aircraft certification model, hardware-in-the-loop result, or physical-flight validation

3.2. Georeferenced disaster-scene construction

The scene layer reads paired pre- and post-disaster tiles, building polygons, damage labels, and geographic metadata derived from xBD. Pixel-to-geographic and geographic-to-pixel affine transforms place each image on its recorded ground footprint. When a tile becomes active, its center defines the local world anchor. Robot, target, and report coordinates remain available as latitude, longitude, and altitude, while local north and east offsets are recomputed relative to that anchor.

The interactive map can place the active disaster image over a satellite or street basemap, display the tile boundary, and color annotated buildings for operator reference. These annotation overlays are not passed to the evaluated agent as visual input. In rendered observations, georeferenced disaster tiles supply task evidence and the surrounding basemap supplies spatial context when a wider footprint extends beyond the available disaster tile. Reference answers and building-level scores remain restricted to the annotated target ROI, so contextual background is not treated as disaster ground truth.

The evaluated scene population contains three disaster events and 44 target ROIs. This is the experimental subset, not a claim that the software can reproduce every disaster represented by the source dataset. Missing tiles, sparse geographic coverage, affine-registration error, and differences in acquisition date all constrain the resulting environment.

3.3. Observation renderer and camera ladder

The simulated sensor is nadir-facing, with horizontal field of view θ and fixed output width W . For vehicle height h , the ground-footprint width L and effective ground sampling distance g are

$$L(h) = 2h \tan(\theta/2), \quad g(h) = \frac{L(h)}{W}. \quad (1)$$

We use $\theta = 60^\circ$, $W = 1024$, and an assumed source scale of 0.5 m/pixel. A 1024-pixel source tile therefore spans 512 m. Footprints of three, two, and one tile widths define cruise, intermediate, and floor states at approximately 1330.2, 886.8, and 443.4 m, with effective GSDs of 1.5, 1.0, and 0.5 m/pixel (Fig. 2). The heights are derived parameters of this camera abstraction; they are not measured flight altitudes from the source imagery.

For paired damage perception, the post-disaster channel is sampled at the current footprint and fixed output size. The corresponding pre-disaster window is retained at the assumed source scale before the pair is brought to the detector input geometry. This represents access to an archival reference and a controllable current observation. It also means the two channels

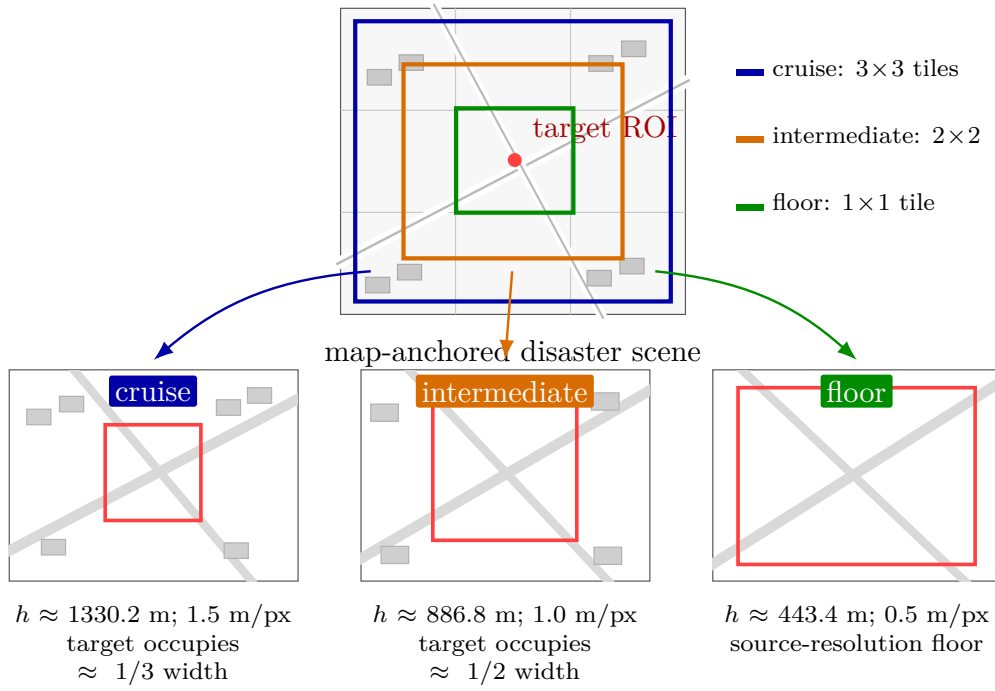


Figure 2: Observation construction from one georeferenced scene. Each state is rendered to 1024×1024 pixels. Descending narrows the ground footprint and assigns more existing source pixels to the target ROI; the floor state reaches, but cannot exceed, the assumed source resolution.

undergo different resampling. The platform records this choice because it can contribute to any observed view response.

Changing height only crops and resamples existing orthographic imagery. The renderer does not create sub-source detail, oblique views, three-dimensional parallax, self-occlusion, atmospheric effects, motion blur, or a new acquisition time. At the floor, further descent cannot add information under the adopted source model.

3.4. Vehicle state and action execution

The world model stores a single UAV with geographic position, altitude, heading, speed, flight status, and task state. It also stores targets, reports, the active disaster tile, map bounds, and timestamps. Motion is implemented by straight-line interpolation in a locally anchored horizontal frame. A position update is converted between geographic coordinates and local north-east offsets, and altitude is recorded as the vertical state.

The task planner and closed-loop agent share a finite set of environment tools: fly to a geographic coordinate, move by a relative horizontal or vertical offset, hover, run disaster detection, mark a target, and produce a report. The inspection controller additionally uses answer, abstain, and stop as episode decisions. A reobservation maps to one or more movement primitives, such as centering on evidence and descending by one camera-ladder state, followed by a new render and perception pass.

This action model supports repeatable studies of where the sensor is pointed and how often another observation is requested. It omits rigid-body attitude dynamics, acceleration limits, wind, terrain following, collision checking, communications, and battery discharge. Executed action count is therefore the principal online cost in this paper. Any time estimate obtained from configured speeds remains a simulator quantity rather than measured endurance.

3.5. Interactive operation and experiment instrumentation

The console exposes disaster selection, a georeferenced situation map, manual flight controls, language-task submission, perception displays, agent state, and event logs. An operator can activate a disaster tile, select a target location, issue an inspection task, observe the planned or closed-loop actions, and inspect the returned damage evidence. The operator console supports qualitative inspection of the same scene state, perception evidence, decisions, actions, budgets, and event logs used by the headless benchmark. The interface is intended for monitoring and diagnosis; all quantitative results in this study are produced by the frozen headless protocol.

The headless benchmark initializes a question and vehicle state, invokes the same observation and action modules, and writes the evolving answer, evidence, controller decision, movement, budget, and terminal outcome. Frozen question identifiers allow policies to be compared on paired tasks. The implementation also supports a fixed-view mode that holds the environment state constant while repeating answer generation. The final benchmark binds generation seeds to question, repeat, step, and call role, making environmental changes and subsequent generation calls separately traceable. Policies may still make different numbers of later calls, so whole-policy comparisons are not pure action-only effects.

4. Closed-loop inspection and observation rules

4.1. Task, state, and evidence

Each inspection episode has a question q , a geographic region of interest (ROI), a task type (presence, damage, count, or spatial relation), and a reference answer y_q . At step t , the simplified vehicle state $s_t = (x_t, y_t, h_t)$ determines a rendered image $I_t = \mathcal{R}(\mathcal{M}, s_t)$ from scene $\mathcal{M}$. The agent also receives structured evidence z_t from the fixed perception backend and registered environment state. It may search, request a task-appropriate reobservation, answer, abstain, or stop. A reobservation is counted only when a movement executes and a new view is rendered; a repeated answer on an unchanged view is not an observation action.

Figure 3 summarizes the agent loop. The benchmark retains the intermediate answer, decision, movement, evidence, generation-call keys, and terminal outcome. It scores the terminal normalized answer against y_q , with abstentions counted as unsuccessful answers. This is an episode-level question-answering endpoint rather than an average of building classifications.

4.2. Fixed binary perception and cross-view memory

The main experiments use the same frozen ChangeOS ResNet-34 checkpoint for every policy. It detects buildings in registered pre-/post-disaster imagery and supplies a binary no-damage/damaged probability vector. ChangeOS is an external perception tool, not a newly trained contribution of this study. Its checkpoint has prior xBD exposure; neither the perception evaluation nor the online results demonstrate transfer of the detector to unseen disaster events.

For a binary vector $p_i = (p_{i0}, p_{i1})$, the damage-uncertainty score is

$$H_2(p_i) = -\frac{1}{\log 2} \sum_{j=0}^1 p_{ij} \log p_{ij}. \quad (2)$$

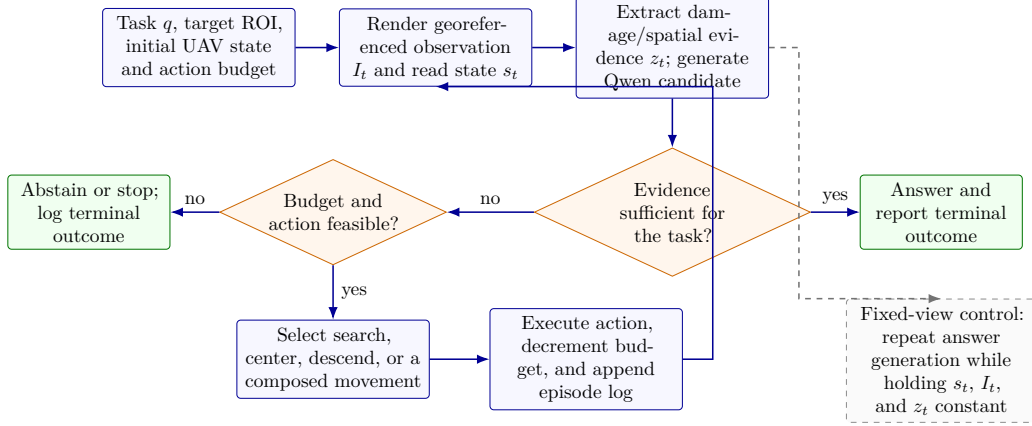


Figure 3: Closed-loop inspection state machine. Feasibility and budget checks distinguish a requested reobservation from an executed view change. The unchanged-view branch is available for diagnosis but is not counted as reobservation.

The raw-entropy policy uses the uncalibrated binary probabilities, whereas calibrated entropy and a prediction-set trigger are supplemental comparisons. Historical four-class temperature and APS thresholds are not transferred to this experiment. For count tasks, the task-conditioned gate estimates uncertainty over the answer buckets 0, 1, 2, 3+ from the binary building probabilities; for spatial tasks it uses registered target geometry and a GSD-dependent bearing uncertainty. Thus a single uncertain building label is not treated as a sufficient reason to crop away a complete count or spatial ROI.

The online evidence memory matches predicted objects by geographic proximity and combines observations across views, preserving the history of observation IDs and ROI coverage. This avoids treating each new crop as an unrelated scene or automatically discarding earlier wide-view evidence. Answer generation uses a locally loaded Qwen2.5-VL-7B-Instruct model with the task image and selected structured state; the evaluated configurations use the same hybrid answer path. The model’s output is normalized to the task’s answer vocabulary before scoring. A deterministic rule answerer and pure-VLM paths exist as development diagnostics, but are not pooled into the seven-policy final comparison.

4.3. Task-conditioned gate and comparators

The task-conditioned policy (T1_TASK) estimates whether an executable next view is worth its cost. In its implemented rule, the candidate utility is

$$U_q = u_q(1 - h_{t+1}/h_t) - 0.05 \hat{T}_{\text{move}}/60 - (1 - \hat{C}_{\text{ROI}}), \quad (3)$$

where u_q is task-specific uncertainty, $\hat{T}_{\text{move}}$ is a configured-speed motion-time estimate in seconds, and $\hat{C}_{\text{ROI}}$ is predicted ROI coverage after the candidate action. This is a fixed heuristic, not a learned or calibrated estimate of expected task accuracy. The frozen gate requires $u_q \geq 0.5$ and $U_q \geq 0.05$; count and spatial tasks additionally require predicted coverage of at least 0.98. A damage question may center on a visible marked target and descend. Count and spatial questions retain the geographic center and descend only if the task ROI remains covered. Presence positives are answered from the wide view, while missing evidence uses the search path. All thresholds were chosen before the final run using development evidence.

The four primary comparators are A0_HOLD (search allowed, no reobservation), A1_RANDOM (independent seeded random trigger), A2_ALWAYS (request whenever feasible), and A3U_RAW_ENTROPY (binary raw-entropy trigger). A3_ENTROPY and A4_CONFORMAL are supplemental. Each active policy has the same exogenous cap of two reobservations per episode, and all configurations permit up to six search steps. A0_HOLD has a zero reobservation cap. The cap is shared, but realized action counts differ because policy requests and feasibility differ; the seven outcomes are operating points, not a randomized budget sweep.

5. Experimental protocol

5.1. Development diagnostics and frozen final tasks

The ChangeOS camera-ladder diagnostic uses 43 development ROIs containing 7,238 annotated buildings. It compares cruise, intermediate, and floor predictions on the same ground-truth building identities. The full-ground-truth endpoint counts an unmatched building as incorrect; matched-only classification and probability metrics use only matched detections. These are development and mechanism measurements, not final policy scores.

The confirmatory online question set is separate from the 160-question development set used to choose policy rules. It contains 160 questions from 44 previously unconsumed ROIs: 48 in Hurricane Michael, 51 in the Moore tornado, and 61 in Nepal flooding. Questions cover presence, damage, count, and spatial relations. Two annotators independently reviewed all 160 final questions before the online experiment. The review checked task type, target ROI, reference answer, and whether the question was answerable from the registered imagery and associated ground-truth evidence. Previously flagged ambiguity notices were retained in the audit trail rather than deleted, and disagreements were resolved before the final set was frozen. The final question set, dual-review record, adjudication log, binary-label manifest, and

source/prompt/weight hashes were checked before the first online episode. The exact identities and checksums are listed in Appendix A.

5.2. *Online execution and endpoints*

Seven policies run on each of the 160 final questions for each of three generation repeats, producing 3,360 planned episodes. Four shards partition each repeat without changing the question set. The action base seed is 42; generation calls derive their seed from base 42000 and (qid, repeat, step, call role). Qwen sampling settings, source fingerprint, prompt hash, and ChangeOS weights are frozen in the protocol. All 12 shard audits passed with zero execution errors. Repeats provide three generation conditions on the same 160 questions; they are not new independent questions, ROIs, or events.

The prespecified primary endpoint is terminal task accuracy over all questions, averaged over repeats. Abstentions count as incorrect for this endpoint. Secondary descriptions include accuracy by event and task type, abstention, within-policy answer correction and harm, executed reobservations, state-derived horizontal/vertical motion, and episode wall time. A correction or harm observed inside one policy’s trajectory is not the counterfactual effect of its action relative to another policy. Later VLM call counts can also differ across policies, so paired accuracy contrasts estimate whole-policy outcomes rather than a pure image-change effect.

Four T1_TASK contrasts against A0_HOLD, A1_RANDOM, A2_ALWAYS, and A3U_RAW_ENTROPY were declared before execution. Pairing uses the same (repeat, qid) units. For each contrast, a 5,000-draw ROI-cluster bootstrap resamples all questions and three repeats within each sampled ROI, with seed 42. The four primary comparisons use Bonferroni-adjusted 98.75% percentile intervals; 95% intervals are descriptive. Only three events are represented, so these intervals characterize variation among the observed ROI clusters, not transfer to unseen disasters. Event and question-type breakdowns are descriptive rather than additional confirmatory tests.

Executed reobservation count is the principal observed action-cost measure. Episode wall time is measured software runtime for this execution, not UAV flight time. Trajectory-state displacements are geometric estimates, and configured-speed d/v values are estimates rather than measured airborne duration. No randomized common-cap sweep or component-level render/ChangeOS/controller/Qwen timing is available. Accordingly, accuracy and cost are reported as realized policy operating points.

6. Results and discussion

6.1. RQ1: A narrower observation changes both detection and correctness

The development camera-ladder diagnostic keeps 7,238 ground-truth buildings in 43 ROIs fixed while changing the rendered footprint. ChangeOS localization recall rises from 22.09% at cruise to 34.06% at intermediate and 56.52% at floor. When unmatched ground-truth buildings are counted as incorrect, binary accuracy rises from 20.01% to 31.06% and 53.65%, respectively. Among matched buildings alone, accuracy is 90.56%, 91.20%, and 94.92%, with macro-F1 of 0.8714, 0.8743, and 0.9232. Matched-only Brier score falls from 0.0657 to 0.0384 between cruise and floor; these probability metrics are not defined over unmatched ground-truth buildings. The large gap between all-GT and matched-only accuracy shows that localization and observation coverage, not just binary classification, dominate the full-building endpoint. Because matched populations change by view, their classification percentages must not replace the fixed-denominator measure.

For the paired cruise-to-floor comparison, 2,696 building outcomes change from incorrect to correct and 261 change from correct to incorrect, a net gain of 2,435 on this development population. The same nominally closer view therefore supplies useful detail and harmful changes. These are binary ChangeOS diagnostics; the archived four-class results in Appendix B belong to a different backend and are not pooled with them. More importantly, building-level gains do not by themselves imply better task answers.

6.2. RQ2 and RQ3: Frozen online policies and task answers

Table 2 reports the complete online comparison. T1_TASK has the highest observed accuracy, 75.00%, compared with 72.29% for A0_HOLD. It uses 161 executed reobservations across 480 question-repeat episodes, compared with 419 for random, 923 for always, and 626 for raw entropy. The higher observed score and lower realized reobservation count are useful descriptive results. They are not a matched-cost comparison: policies have the same maximum cap but choose different numbers of actions.

The four prespecified paired contrasts are in Table 3. T1_TASK exceeds A0_HOLD by an observed 2.71 percentage points, but the multiplicity-adjusted ROI-cluster interval includes zero. The appropriate claim is that T1 is higher *in this final run*, while a stable advantage over holding remains uncertain. Its contrast with raw entropy is likewise uncertain. The interval against random touches zero at its lower endpoint, so it does not establish a strict positive difference under the declared interval rule. T1 is above always reobserving within the prespecified cluster analysis, but this comparison alone does not prove that its individual movements caused the gain.

Table 2: Frozen final-set online policies. Each row contains the same 160 questions under three generation repeats (480 question-repeat episodes); repeats are not independent questions. Wall time is measured software episode runtime, not flight duration. A3 and A4 are supplemental comparisons.

Policy	Correct / 480	Accuracy (%)	Reobservations	Wall time (s/episode)
A0_HOLD	347	72.29	0	17.03
A1_RANDOM	326	67.92	419	33.65
A2_ALWAYS	307	63.96	923	51.07
A3U_RAW_ENTROPY	339	70.63	626	39.61
A3_ENTROPY	341	71.04	628	38.94
A4_CONFORMAL	338	70.42	628	38.95
T1_TASK	360	75.00	161	17.92

Table 3: Prespecified T1_TASK paired accuracy contrasts. Differences are percentage points; 98.75% ROI-cluster percentile intervals adjust for the four primary comparisons. All questions and repeats from a sampled ROI remain together.

Contrast	Difference (pp)	98.75% interval (pp)
T1 minus A0_HOLD	+2.71	[−2.04, +7.50]
T1 minus A1_RANDOM	+7.08	[0.00, +14.70]
T1 minus A2_ALWAYS	+11.04	[+0.86, +21.38]
T1 minus A3U_RAW_ENTROPY	+4.38	[−4.33, +12.96]

Additional observations can alter the answer in either direction. Within their own episode histories, T1_TASK records 36 corrected and 21 harmed answers; A2_ALWAYS records 74 corrected and 132 harmed. These transition labels are not paired counterfactual effects versus A0_HOLD. They nevertheless show why an always-reobserve policy can spend the largest action budget while finishing with the lowest accuracy. A1_RANDOM executes 419 reobservations and reaches 67.92%; A3U_RAW_ENTROPY executes 626 and reaches 70.63%. Neither action count nor classifier uncertainty alone orders final task quality.

The episode traces show concrete instances in both directions. In the first generation repeat, a Hurricane Michael damage question (ID ending `damage_0001_9430`) changes from an initially incorrect “no damage” answer to the correct “damaged” answer after two T1 reobservations. A Hurricane Michael spatial question (ID ending `spatial_0092_5452`) changes from the correct “north” to the incorrect “southwest” after one. These examples describe recorded answer sequences, not isolated causal effects of the movement.

6.3. RQ4 and RQ5: Task dependence, bottlenecks, and scope of the gain

Task-type outcomes reveal uneven effects. For T1_TASK versus A0_HOLD, damage accuracy is 75.76% versus 66.67% and count accuracy is 78.29% versus 75.97%. Presence is identical at 93.18%; spatial accuracy is lower for T1, 41.38% versus 43.68%. These are descriptive, correlated subgroup summaries, not four additional adjusted tests. They are consistent with the design premise that a wide context can already suffice for presence while a more detailed target view may help damage, but the spatial result limits any universal benefit claim.

Event outcomes are also heterogeneous. T1_TASK versus A0_HOLD is 54.86% versus 58.33% in Hurricane Michael, 84.31% versus 77.12% in the Moore tornado, and 83.06% versus 79.23% in Nepal flooding. The pooled numerical gain is not reproduced in every event, and three events cannot support an unseen-disaster generalization claim. Accuracy across T1’s three generation repeats is 75.00%, 74.38%, and 75.63%; repeat stability does not remove ROI or event uncertainty.

Development diagnostics provide limited bottleneck evidence without forming an additive oracle decomposition. A deterministic answerer using predicted structured evidence reaches 66.87% on an earlier development run. Full-ROI ground-truth evidence with a deterministic rule answerer reconstructs 160/160 development reference answers, checking the question/label pipeline rather than online performance. Their source fingerprints and interventions differ, and current-visible GT with Qwen, full-task GT with Qwen, and a full-coverage always-floor reference remain unmeasured. The final

T1/A0 comparison therefore isolates neither the perception error nor the language-generation error as a unique cause.

The observed T1 operating point combines a modest numerical accuracy gain over holding with fewer executed reobservations than other active policies. Its uncertainty interval, event variation, and incomplete oracle chain favor a platform-and-mechanism interpretation rather than a demonstrated generally superior task-conditioned algorithm. A randomized common-budget sweep would be needed to estimate a genuine accuracy–cost curve; the seven realized operating points cannot supply one by rebinning episodes after execution.

7. Limitations and deployment boundary

The camera ladder crops and resamples existing orthographic imagery. It changes geographic footprint and assigned source pixels, but it cannot synthesize a new viewing angle, acquisition time, illumination condition, occluded surface, or detail below native resolution. Pre- and post-disaster channels undergo different resampling, and basemap context may come from a different date. The fixed flat-ground, nadir abstraction does not validate altitude-dependent image formation for a physical UAV.

The vehicle model uses simplified position updates without wind, attitude dynamics, acceleration limits, terrain and collision avoidance, communications, or battery discharge. Executed reobservations are measured software actions. State-derived horizontal and vertical displacements are geometric estimates. Of 3,360 final episodes, 42 end with a fly action lacking a subsequent recorded state, so complete-motion summaries are available for 3,318. Episode wall time is measured execution time on the experiment system, not measured flight time; render, ChangeOS, controller, and Qwen component latencies were not separately recorded. The study therefore concerns imagery-grounded active observation, not operational flight performance.

The ChangeOS checkpoint is fixed across policies but has prior exposure to related xBD data; the final policy comparison does not imply event-disjoint perception training. The final questions occupy 44 ROIs in only three disaster events. All 160 final questions were independently reviewed by two annotators and adjudicated before the set was frozen. This reduces reference-answer and task-scope ambiguity, but it does not eliminate possible dataset artifacts. Answer frequencies and template families may still permit language-only shortcuts, and the present study does not include a separate language-only evaluation on the frozen final set.

The generation protocol gives three seeded repeats, but active policies can make different numbers of later Qwen calls. Their paired outcomes thus compare entire decision pipelines rather than the isolated causal effect of

one movement. The seven policies share a maximum reobservation cap, not equal executed action counts. No randomized cap sweep, full module-level timing, or complete layered GT/Qwen oracle suite is available. The interval for T1_TASK versus A0_HOLD includes both a small disadvantage and a useful advantage, and one of three event directions is negative. These boundaries preclude claiming stable superiority over holding or generalization to unseen disasters from the present sample.

Archived four-class allocation and VLN component results in Appendix B use different populations and/or an earlier observation stack. They are historical diagnostics, not replications of the binary ChangeOS final experiment, and their calibration constants or performance numbers are not transferred into the main evidence chain.

8. Conclusion

DisasterClaw turns georeferenced pre- and post-disaster imagery into traceable closed-loop inspection episodes. A fixed binary ChangeOS detector, simulated observation control, task-conditioned action gate, evidence memory, and question-answering interface make it possible to follow a sensing decision from movement through perception to the terminal answer.

In 160 dual-reviewed final questions over three generation repeats, T1_TASK reaches 75.00% accuracy versus 72.29% for holding and executes fewer reobservations than the other active policies. This is an observed improvement in the tested sample, not an established stable advantage over no reobservation: the prespecified adjusted interval for the 2.71-point difference spans zero. The always-reobserve policy uses 923 actions and finishes at 63.96%, illustrating that more observation can also be harmful. Together with the development camera-ladder’s corrections and harms, these results support the platform’s role in measuring task-dependent observation value.

The next scientific steps are a randomized common-budget comparison, fuller oracle decomposition, component-level timing, independently reviewed questions across more events, and a perception checkpoint evaluated without event exposure. Claims about real flight require external action and imaging validation. Until then, the contribution is an imagery-grounded environment and a measured policy comparison with explicitly bounded evidence, not a demonstrated general-purpose flight policy.

Declarations for author completion

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CRedit author statement. *To be assigned and approved by the named authors.*

Declaration of competing interest. *To be confirmed by all authors.*

Data and code availability. The working package contains manuscript source, frozen P6 protocol and hashes, audited final episode logs, and analysis scripts. Access to the underlying imagery remains subject to its original terms. A public archive, license, anonymization review, and complete imagery/checkpoint package have not yet been confirmed; a final availability statement must identify the actual release and its restrictions.

Generative AI assistance. Generative AI tools were used to assist manuscript restructuring, English-language editing, reference checking, and preparation of analysis and typesetting scripts. All scientific claims, experimental design, reported results, and final manuscript content were reviewed and approved by the authors, who take full responsibility for the work.

Appendix A. Reproducibility and provenance

Appendix A.1. Frozen final identities and audit boundary

The final protocol is stored in `cja_en/review/changeos_p6_final_protocol_20260915.json`. It binds the dual-reviewed task set, review and adjudication records, frozen three-event manifest, ChangeOS weights, source fingerprint, system prompt, policy configuration, and analysis scripts by full SHA-256 values. Its own SHA-256 begins `41ccbb5fed5e`; exact paths and full hashes are retained in the machine-readable protocol rather than truncated in the audit. These identities describe the local execution; they are not substitutes for a publicly recoverable image and checkpoint package.

The final task package also contains the two independent review records and the adjudication record used to freeze the 160-question set. Review artifacts are linked to the frozen question identifiers and are retained as provenance rather than being used as model inputs.

The 12 final shard directories contain manifests, preflight receipts, `episodes.jsonl`, `results.json`, and strict per-shard audits. Their combined report is `p6_final_three_event_v1_20260915_report.json` under `runs/benchmarks/changeos_binary/`; its SHA-256 begins `9e4370325bc9`. It validates 3,360 rows with zero execution errors, exact question/configuration/repeat coverage, and declared input identities. This is a local execution and consistency audit, not an independent re-execution of the full model pipeline. The run uses a fixed ChangeOS checkpoint and a locally loaded Qwen2.5-VL-7B-Instruct model; checkpoint identity and generation settings must accompany any release rather than be inferred from an online model name.

Scene construction and rendering are in the `backend/` modules `world.py`, `geo.py`, `xbd_map.py`, `fov_ladder.py`, and `perception.py`. Inspection, reobservation, and local Qwen calls are in the same directory’s `agent_vqa.py`, `recheck.py`, and `local_qwen_vl.py`. The headless protocol is under `scripts/benchmarks/`; the operator interface is under `frontend/src/`. This file map establishes software correspondence, not that every optional module was active in each policy.

The final cost audits are `runs/benchmarks/changeos_binary/p7_cost_observability_final_repeat0_20260917/report.json`, with corresponding `repeat1` and `repeat2` paths. The existing auditor was applied to four shards at a time because its duplicate key is not repeat-aware. Together the three reports cover 3,360 episodes and flag 42 incomplete terminal-motion traces. All logged recheck request, inferred allocation, execution, and episode totals agree. Search-action allocation details, component-level timing, actual flight time, and physical action mapping remain unavailable. A complete public reproduction package would additionally need the licensed source imagery, scene manifests and affine transforms, basemap availability, dependency lock, checkpoint retrieval instructions, and one replay with action/observation hashes.

Appendix B. Historical diagnostics outside the main evidence chain

Appendix B.1. Archived four-class camera and allocation study

An earlier `xview2_first` four-class pipeline evaluated 3,753 buildings in 44 ROIs. It is not the binary ChangeOS backend of the final study. Table B.1 retains its camera-footprint results and Table B.2 its offline allocation at a 0.25 building-selection fraction. The old temperature, prediction-set threshold, four-class entropy normalization, and allocation scores are not used in the main experiment. These offline selections operate on paired stored predictions, not on executable closed-loop routes or question answers.

Table B.1: Historical four-class building-level camera diagnostic on the same 3,753 evaluation buildings. Not comparable to the binary ChangeOS final question endpoint.

View	GSD (m/px)	Acc.	Macro-F1	Brier	NLL
Cruise	1.50	0.7250	0.6118	0.4814	1.3564
Mid	1.00	0.7357	0.6327	0.4669	1.3314
Floor	0.50	0.7423	0.6542	0.4616	1.3073

The archived matched-only sensitivity population contains 2,757 buildings detected at all three views (Table B.3). Because the subset is selected by

Table B.2: Historical four-class offline allocation at a 0.25 building-selection fraction. The label-informed row uses evaluation labels and is not deployable.

Policy	Actions	Acc.	Macro-F1	ECE	Net correct
No reobservation	0	0.7250	0.6118	0.2423	0
Random	938	0.7301	0.6241	0.2437	19
Raw entropy	938	0.7397	0.6448	0.2578	55
Calibrated entropy	938	0.7391	0.6433	0.2609	53
Expected entropy reduction	938	0.7357	0.6308	0.2551	40
APS set size	938	0.7343	0.6367	0.2549	35
Label-informed diagnostic	938	0.7759	0.6928	0.2853	191

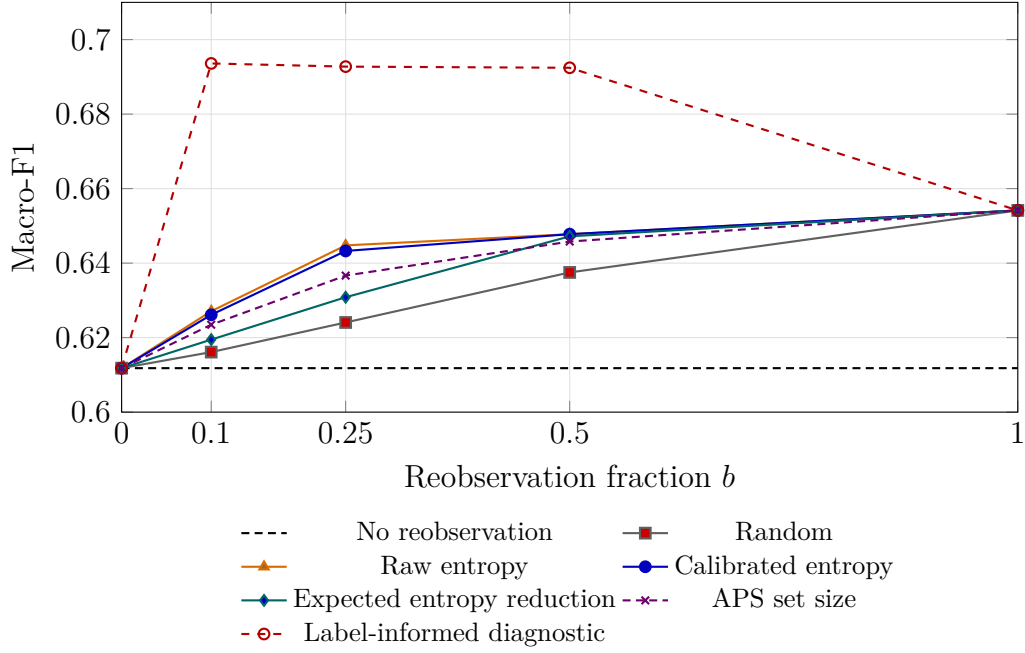


Figure B.4: Historical offline four-class allocation sweep. Selected floor predictions replace cruise predictions; this is not a common-budget online policy curve.

detector success, it cannot replace the full 3,753-building endpoint. Its earlier exploratory ROI-bootstrap intervals, shown in Table B.4, condition on fixed scores and the observed three events; they do not demonstrate transfer to new disasters.

Table B.3: Historical four-class matched-only sensitivity analysis.

View	Accuracy	Macro-F1	Brier	NLL
Cruise	0.7918	0.7069	0.3367	0.6455
Mid	0.8052	0.7321	0.3176	0.6123
Floor	0.8099	0.7489	0.3175	0.6103

Table B.4: Historical four-class exploratory paired ROI-bootstrap differences at the 0.25 offline allocation fraction.

Paired comparison	Macro-F1 difference	95% interval
Calibrated entropy – Random	+0.0192	[+0.0097, +0.0419]
Calibrated entropy – Raw entropy	-0.0015	[-0.0093, +0.0061]

Appendix B.2. Archived VLN switch diagnostic

The earlier 200-instruction VLN switch experiment used an earlier observation stack and one recorded run per configuration. Table B.5 preserves the archived aggregate as a component diagnostic only. Enabling the recheck configuration did not cause recheck actions in that run; the table cannot establish a benefit from planning, memory, or reobservation under the current ChangeOS binary protocol.

Table B.5: Archived VLN switch diagnostic. SR is strict success within 25 m of the specified target; semSR accepts a matching damage-class building within 25 m. Recheck triggers were zero in all four configurations.

Config.	Enabled navigation components	n	SR (95% CI)	semSR	NE (m)
B0	Keyword/greedy baseline	200	0.070 [0.042, 0.114]	0.165	222.3
B1	Hierarchical semantic planning	200	0.080 [0.050, 0.126]	0.155	223.9
B2	B1 + uncertainty recheck	200	0.080 [0.050, 0.126]	0.155	223.9
B3	B2 + topological memory	200	0.080 [0.050, 0.126]	0.140	305.7

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